

Accounting for Italy’s North–South divide in trusting behaviour: Experimental evidence from Southern migrants*

Gianluca Grimalda[†] Andrea Guido[‡]

July 22, 2026

Abstract

Regional economic divides often persist despite free mobility, a puzzle attributed to gaps in trust and cooperation preferences. We study Italy’s North–South trust divide to test whether such differences dissolve after migration to a higher-trust region. Using trust-game data from a Northern Italian province, we compare natives, Southern migrants, and other-region migrants: Southern migrants trust 29% less than natives, while other migrants are indistinguishable from them. The gap reflects both more pessimistic beliefs about others’ cooperation and lower trustworthiness, with beliefs playing the larger role. Trust does not converge toward natives’ levels with time since migration, suggesting persistent cultural traits.

JEL codes: C91, D91, Z13, R23

Keywords: Trust; Migration; Social Capital; Beliefs; Experimental Economics; Mediation Analysis

*This paper stems from a research project on “Trust and Trustworthiness in Association Members and in the Society.” Two papers were derived from the project; this is a complementary work to [Degli Antoni and Grimalda \(2016\)](#) and [Degli Antoni and Grimalda \(2024\)](#). The project was funded by the following grants: Bancaixa P1-1A2010-17 and P1-1B2010-17, Ministerio de Ciencia e Innovación de España (ECO2011-23634), Junta de Andalucía (P07-SEJ-03155), and the Italian Ministry of University and Research under the national research project (PRIN) n. 20085BHY5T.

[†]Masaryk University.

[‡]Paris School of Business, Paris, France.

1 Introduction

Many regions within the same country experience large disparities in economic outcomes. This is puzzling because, in the absence of any institutional, language, and legal constraints, capital mobility within a country should bring about economic convergence. A burgeoning strand of literature has put forward an explanation based on beliefs and cooperation preferences differences as a possible explanation for such regional divides (Bigoni et al., 2016, 2019). Long-standing cultural differences have led to heterogeneous beliefs and preferences for cooperation, spanning decades (Alesina and Fuchs-Schündeln, 2007) or even centuries (Putnam et al., 1993; Banfield, 1958)¹, a crucial aspect for economic development (Arrow, 1974; Knack and Keefer, 1997; Guiso et al., 2004; Tabellini, 2010). Such behavioral gaps may thus create a specific kind of poverty trap, where distrust in others’ cooperation thwarts both interpersonal cooperation and economic growth.²

While substantial work has been devoted to shed light on the historical origins of differences in social preferences and beliefs across cultures (Henrich et al., 2005)³, there is a general lack of understanding on whether these differences adapt or persist over time when members of a culture move to a different one. Policymakers often focus on devising strategies that promote integration, believing it fosters societal outcomes. In this paper we shed light on the adaptability of trust and trustworthiness, as well as on the factors that can hamper or facilitate such adaptation. We draw upon the Italy’s North–South divide in trust (Bigoni et al., 2016) to conduct an experiment with an adult sample of residents and migrants in one of Italy’s regions with the highest level of social capital.

The case of Italian migration lends itself well to studying the gap in social preferences and beliefs across regions, because people migrating from a low-trust region to a high-trust region may, in principle, quickly adapt to the new context and adopt the predominant social norms.

¹More precisely, Putnam et al. (1993) see the establishment of city-states in Italy’s North during the High Mediaeval Period (c. 1001–1300) as the major trigger of a “civic culture” favoring co-operation, mutual trust and reciprocity in Italy’s North. On the contrary, the feudal institutions characteristic of the absolutistic monarchy in the South prevented a similar development. See also (Guiso et al., 2008; Sabatini, 2008) for confirmation of this thesis. The idea that cultural differences spawned by past political institutions help explain current economic outcomes has also been generalised to account for differences across European regions (Tabellini, 2010).

²See Fischbacher et al. (2001) for a taxonomy of behavioral types in cooperation interactions. Since most individuals are conditional cooperators, i.e., are willing to cooperate as long as they expect enough others to cooperate, beliefs in others’ cooperation will have a key role in determining cooperation levels in a group. Beraldo (2010) and Daniele and Malanima (2013) claim that the Italian North–South divide is an instance of a “poverty trap” due to economic structural conditions.

³The seminal book by Putnam et al. (1993) pointed to cultural factors, and the related historical determinants, as the origin of the century-long wide economic gap between Italy’s Northern and Southern regions. This thesis has received further support in more recent analyses (Guiso et al., 2008; Sabatini, 2008).

Understanding whether the lack of cooperation in regions characterized by interpersonal distrust is due to differences in beliefs or social preferences is of topical interest for policy makers, as it has been argued that the former are more malleable to change than the latter (Bigoni et al., 2019; Thöni et al., 2012).

We draw upon experimental data on trust and trustworthiness in a Trust Game (Berg et al., 1995) from a stratified adult sample of residents of Emilia Romagna, one of the regions with the highest level of interpersonal trust in Italy (Putnam et al., 1993; Baldassarri, 2020). This work stems from previous work that compared trust behavior between individuals belonging to cultural associations and non-members (Degli Antoni and Grimalda, 2016). In this follow up, we use the fact that approximately 20% of our sample comprises people who were born in the South of Italy to analyse differences prosocial behavior across three groups: immigrants from Italy’s South – an area characterized by low levels of interpersonal trust (Putnam et al., 1993; Bigoni et al., 2016; Tabellini, 2010) – immigrants from other Italian regions, and natives to Emilia Romagna. Moreover, for a portion of our migrant sample we are able to measure the length of stay in the North since migrating. Relocation to the region ranged from less than a year to virtually one’s whole life, thus offering high variation in one’s exposure to the new environment. This data enable us to address the following questions: (a) Are differences in trust and trustworthiness between Southern migrants and other groups persistent after migration? (b) Do levels of trust and trustworthiness between natives and immigrants converge over time as immigrants spend longer spells in the region? (c) If behavioral differences persist, can these be related to individual preferences – such as social preferences or risk aversion, or beliefs about others’ behavior?

The first two questions shed light on the adaptability of individual behavior after a considerable change in their social environment has taken place. How quick are individuals to adapt to an environment in which trust, rather than withdrawing trust, is the payoff-maximizing strategy? In spite of the relevance of the topic, only very few contributions have focused on how preferences and cultural traits evolve over time when people migrate to another country or region (Cameron et al., 2015; Guillén and Ji, 2011; Cox and Orman, 2015). On this end, the experimental literature provides mixed evidence. Guillén and Ji (2011) find that Asian international students who migrated to Australia display overall similar levels of trust towards both other international students and domestic ones. However, trust levels decrease as migrant students spend more semesters in the hosting country. Same behavioral patterns are found also in Cameron et al. (2015), using a similar sample of subjects. Differently, Cox and Orman (2015) find that the length of time an immigrant has been a naturalized U.S. citizen has no impact on trusting behavior, but positively affects level of trustworthiness.

Our work complements the burgeoning literature that has compared cooperative behavior of people from the North and South of Italy. In [Bigoni et al. \(2016\)](#), people residents in Southern Italian locations interacting with other Southerners display lower cooperation levels than Northerners interacting with Northerners in the context of both Public Goods games and Trust Games. In a follow-up study, [Bigoni et al. \(2019\)](#) investigate whether such behavioral gap stems from differences rooted in pro-social preferences or beliefs. The authors find very similar preferences for cooperation between university students from Southern Italy, who resided in Northern Italy at the time of the experiment, and natives from the North. However, both Southerners and Northerners correctly predicted lower cooperation among the former than among the latter sample in the previous experiment ([Bigoni et al., 2016](#)). This evidence clearly supports the idea that differences in cooperation in the Italian divide are due to differences in beliefs rather than social preferences.

While the closest work to ours is [Bigoni et al. \(2019\)](#), our study differs from theirs in many respects. First, we find that levels of trust by natives are about 29% higher than those by migrants from the South. Conversely, immigrants from Northern or Central regions of Italy display trust levels that are indistinguishable from natives'. Moreover, both beliefs about others' behavior and social preferences differ between Southern immigrants and other groups. Southerners have both significantly lower beliefs on others' trust and trustworthiness than others and display lower prosociality, too. The latter is measured by the degree of trustworthiness in the Trust Game, which is 34% lower for Southern immigrants compared to natives. We also perform a mediation analysis ([Sobel, 1982](#)) to quantify the incidence of beliefs vis-à-vis pro-social and risk preferences. Both beliefs on others' trust and trustworthiness and social preferences mediate the behavioral difference in trust. A robustness check seems to attribute a larger role to beliefs in others' trust, which points to the importance of social norms in one's motivations ([Bicchieri, 2005](#); [Andrighetto et al., 2026](#)). Finally, we do not find evidence that a longer portion of one's life spent in the region by immigrants is associated with behavior more similar to that of natives. This result may be due to the low sample size of immigrants for which we have information on time since immigration, although we do not find significant differences between the sub-samples for which we have such data and those for which we do not. A robustness check of our results comes from comparing differences in patterns of behavior for association members and non-members. Migrants who belong to an association have arguably had better opportunities to integrate into the new social environment and to learn the corresponding social norms. Nevertheless, we find that the above results hold almost identically for members and non-members. This confirms the resilience of untrusting and untrustworthy behavior by Southerners.

Overall, our results point to strong persistence of cultural traits acquired from the area of birth. It is remarkable that such differences persist throughout one’s life time, even when Southern immigrants migrate to one of the regions with the highest levels of interpersonal trust. This persistence suggests that cultural differences may be hard to change within one’s own lifetime, while they may be more malleable to change across generations (Algan and Cahuc, 2014; Guiso et al., 2004). We can only conjecture that differences between our study and Bigoni et al. (2019) may be due to the relative high cultural homogeneity of university students from different regions.

The paper is organized as follows. Section 2 describes the experimental design and sets out the main hypotheses of our study. Section 3 presents our results. Section 4 discusses the limitations of our results and their generalizability to other contexts.

2 Experimental design

2.1 The Trust Game

Our Trust Game (TG) reproduces Berg et al.’s (1995) seminal design. In our experimental design, a Sender and a Receiver were randomly matched and endowed with 25 Euros (€) each. The Sender had to decide which portion of her endowment to send to the Receiver. The Sender could send any multiple of 5€ from 0€ to 25€, so there were six possible transfer levels. The amount sent was multiplied by two and transferred to the Receiver who then had to decide which portion of the total sum in her possession to send back to the Sender. The Receiver made her decision with the strategy method. She had to report in a form the amount they wished to send back for each of the possible six options available to the Sender. The Receiver could return any amount (up to the first decimal digit) between zero and the sum of the initial endowment of 25€ and the doubled amount transferred by the Sender.

In our experiment, each participant played one TG in the role of Sender and one in the role of Receiver. Pairs were reshuffled after the first decision and subjects were informed that their partner in the second decision was different from the partner in the first decision. Subjects were paid only for one of the two decisions, each having 50% probability of being drawn. No feedback was given at the end of each decision, so we can consider them as independent.

After the two TGs were played, two measures of beliefs were elicited. The first is the belief over how many tokens the Receiver will return, given the Sender’s actual transfer. We asked how much the player expected the Receiver with whom she was paired would send back, given the amount the player actually sent. The second measure is the belief over

how many tokens the Sender will send. Following [Bohnet and Baytelman \(2007\)](#), this belief measure can be interpreted as one’s belief over the prescription of a social norm of “good”, or appropriate, behavior. We asked players to estimate the amount transferred by the Sender with whom they were paired when acting as Receivers. Both measures were monetarily incentivized. Subjects received 1€ for each correct guess, allowing for a $\pm 3\text{€}$ margin of error in the first estimate. Finally, we administered the demographic questionnaire. Payments were distributed by cash at the end of the session. Average payoffs were 31.7€ (std. dev. 11.99). In three cases a participant in the pair earned nothing while the other earned the maximum available – 75€.

Sessions were run in parallel by two experimenters in two different rooms of a library at the University of Parma. Instructions, further details on the experiment protocol, and the questionnaire, can be found in the SOM: section IV.

2.2 Sample characteristics

This work is part of a larger experimental research project aiming to investigate trust and trustworthiness of voluntary association members with respect to non-members.⁴ For this reason, the sample includes association members (i.e. subject who are formally affiliated to an association, and attend association meetings for at least an hour each month), subjects who had never been members of a voluntary association, and subjects who had been members in the past but their associational activity was finished at the time the research was run. The recruitment of association members was carried out in person by experimenters through announcements at association meetings (see the SOM: section II for a detailed description of the associations involved in the research project). For the recruitment of non-members, we relied on an agency (Demoskopoea) that reached subjects through phone calls. It is reasonable to expect that migrants who belong to an association have arguably had better opportunities to integrate into the new social environment and to learn Northerners’ different patterns of behavior than non-members. We will use the condition of belonging to an association as demographic control variable in our regression analyses.⁵

⁴The full description of the project can be found in ([Degli Antoni and Grimalda, 2016, 2024](#)). In there, it has been shown that members have significantly higher trust and trustworthiness than non-members.

⁵A further control variable used in our analyses is the in-group condition that was implemented for the sub-samples of northerners and southerners, as well as members and non-members, in the companion paper ([Degli Antoni and Grimalda, 2016](#)). In the in-group condition, participants were told that: “The person with whom you will be paired is a member of the Association X {researcher stated the name of the association} of which you are also a member, and is resident in Parma, or its province, or in neighboring provinces.” In the General Population condition, subjects were only informed that: “The person with whom you will be paired is resident in the province of Parma or in neighboring provinces.”

A total of 372 subjects participated to the experiment. During recruitment, we requested that subjects had resided for at least one year prior to the research in the province of Parma, where we conducted the experiment, or in neighboring provinces. A total of 59 subjects (16% of the total sample) were born in Southern Italy, 79 subjects (21%) were born in other Italian regions and are classified as Non-Southern Migrants, and the remaining 234 subjects (63%) are Natives (Table 1). We based our definition of South on that used by the Italian Institute for National Statistics (ISTAT).⁶ Subjects coming from central regions (namely, 17 subjects) are placed into either of the two categories, i.e., Non-Southern or Southern migrants.⁷ Table S1 in the Supplementary Online Materials (SOM) reports the demographic characteristics of our sub-samples of people who were born in Southern Italy and Northern Italy. Tests reveals that the sample of Southerners is significantly different from the sample of Northerners with respect to gender composition (Chi Square test: $p = 0.001$) and age (Mann–Whitney test: $p = 0.039$). No differences emerge with respect to education (Chi Square test: $p = 0.871$). This makes the econometric analysis relevant to control for these differences.

3 Results

Our analyses aim at comparing our sample of migrants born in the South of Italy relative to both natives subjects and migrants from northern regions. However, results hold even when excluding migrants from regions other than south of Italy. The variables of main interest in the empirical analysis are:

- The amount sent by the Sender, which we call *Amount sent*. This can be any multiple of 5€ from 0€ to 25€.
- The amount returned by the Receiver. We normalize this variable to the $[0, 1]$ interval by dividing it by the maximum possible amount that Receivers may return. We call the resulting variable *Return rate*. Recall that the Receiver could send back any amount ranging from a minimum of zero up to a maximum given by the sum of

⁶ISTAT classifies the following regions as being part of Italy’s South: Abruzzi, Basilicata, Calabria, Campania, Molise, and Puglia. We also add Sicily to this block, although ISTAT classifies it as belonging to the “Islands” group. Sicily is commonly included in the South of Italy by scientific contributions on trust both for historical and for geographical reasons. In particular, all these regions belonged to the “Kingdom of Two Sicilies,” under the dominion of the Spanish branch of the House of Bourbon, before Italy’s unification in 1861. Merging Sicily with other regions also seems appropriate on the basis of Putnam et al.’s (1993) argument that the existing differences in social capital in Italy between North and South can be tracked down to the regions’ different historical trajectories prior to Italy’s unification.

⁷Among the regions classified as Italy’s Centre, we only consider subjects from the region Lazio as Southerners, consistently to Bigoni et al. (2016).

the 25€ endowment and twice the amount sent to her by the Sender. In formulas, $ReturnRate_A = \frac{Return}{25+2*A}$. Returns were allowed up to the first decimal digit.

- The belief over the Receiver’s return rate, given the Sender’s actual transfer. The elicited variable was the total Receiver’s return, but, analogously to the *Return rate*, we normalize this variable to the $[0, 1]$ interval dividing it by the maximum possible amount that the Receiver may return. We call this variable *Return rate exp.*⁸
- The belief over the amount sent by the Sender, which we call *Amount sent exp.*
- *Southern Migrants*: dummy variable which takes the value of 1 for subjects who were born in Southern Italy.
- *Non-Southern Migrants*: dummy variable which takes the value of 1 for subjects who were born in regions other than Southern Italy and migrated to the north.

3.1 Beliefs about Trust and Trustworthiness

First, we look at the effect of being born in the South of Italy on beliefs. Southerners generally hold more pessimistic beliefs about the amount sent (30% lower than Natives, 34% lower than other migrants) and returned (33% lower than natives, 32% lower than other migrants) than both Natives and non-south migrants (Mann–Whitney test, all $p < 0.001$; Table 1).⁹

To account for the data structure, we fitted an ordered logit model to the expectation of the amount sent by the Sender (eq. 1–2 in the SOM, section I), given the discrete nature of this variable, and a Tobit regression to model Receivers’ beliefs (eq. 2–3 in the SOM, section I).

⁸Note that we did not elicit the returned amount for all the six possible transfer levels, but only for the amount actually sent by the Sender. Eliciting the whole range of beliefs would have of course been interesting. However, we felt that this would have required too long a time and too big a cognitive effort for our adult sample, especially considering belief elicitation occurred towards the end of the session.

⁹These results hold even when considering only participants in associations ($z = 3.412$, $p < 0.001$ for expected returns; $z = 3.001$, $p = 0.003$ for expected amounts sent) and those not participating ($z = 2.283$, $p = 0.022$ for expected returns; $z = 2.840$, $p = 0.004$ for expected amounts sent).

Table 1: Descriptive statistics of experimental decisions across categories

	Amount sent		Avg. return rate		Return rate exp		Amount sent exp	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Southern Migrants	10.08 (7.22) [59]	10	0.231 (0.209) [59]	0.213	0.221 (0.190) [59]	0.182	9.224 (5.604) [59]	10
Non-Southern Migrants	13.670 (6.636) [79]	15	0.310 (0.235) [79]	0.307	0.327 (0.177) [79]	0.285	13.910 (6.429) [79]	15
Natives	14.252 (6.483) [234]	15	0.312 (0.219) [234]	0.307	0.329 (0.171) [234]	0.333	13.197 (6.243) [234]	15

Standard deviations in round brackets; number of subjects in squared brackets.

The effect associated to *Southern Migrants* is significant at all conventional levels and negative in all belief regression models (Table 2). Migrants that are not originally from southern regions of Italy display statistically similar beliefs as natives (variable *Non-Southern Migrants*). Among the demographic controls (see SOM: Table S4), women (variable “Gender”) and people who are particularly dissatisfied with their financial situation (variable “Income_dissat”) have lower expectations over others’ amount sent.¹⁰ Subjects who attend religious services at least once a month have (weakly) higher expectations both over others’ amount sent and over others’ return rate in comparison with people attending religious services less frequently or never (variable “Practicing_Catholic”). Among the control variables, we also notice a significant effect of a dummy variable identifying one of the two experimenters who conducted the sessions (variable “Experimenter”; see SOM: Table S4, columns 1 and 2).¹¹ However, all these effects disappear when controlling for the amount sent (columns 3–4). We then conclude:

¹⁰However, these effects are not significant in model specifications 3 and 4 when controlling for the amount sent, Table S4 in the SOM.

¹¹We were surprised to find this effect, because both experimenters strictly followed the same written protocol and gave instructions according to an identical written script (SOM: section IV). The experimenters had observed each other while conducting a pilot session prior to the beginning of the research, in order to level out differences in their style in conducting the sessions. An external observer had found no relevant differences between the two experimenters’ conducting styles in the pilot. To check whether the results reported above are caused by a specific experimenter effect, we interact this dummy variable with *Southern Migrants*. We find no significant effects of such interacted terms in predicting the two expectations measures. This indicates that the result we found for *Southern Migrants* should not be driven by which experimenter conducted the session. Moreover, we also note that *Experimenter* becomes not significant once expectations measures are included in the model (see SOM: models 3–4, Tables S4 and S5).

Result 1: People born in the South of Italy expect their counterpart to return significantly less when they act as Senders and to send significantly less when they act as Receivers than people born in the North.

Table 2: Regression of beliefs on Amount Sent and Returned

Dependent variable	(1) Amount sent exp	(2) Amount sent exp	(3) Return rate exp	(4) Return rate exp
Southern Migrants	−1.012*** (0.299)	−0.798* (0.363)	−0.126*** (0.032)	−0.087* (0.036)
Non-Southern Migrants	0.180 (0.262)	0.133 (0.259)	−0.010 (0.027)	−0.006 (0.026)
Amount Sent		0.222*** (0.022)		
Average Return rate				0.349*** (0.078)
Demographic and additional controls	Yes	Yes	Yes	Yes
Observations	317	317	317	317

Standard errors in parentheses. 55 out of the 372 subjects were excluded because of missing values in the controls implemented. Results are unchanged also without controls (see SOM). Columns (1)–(2): ordered logit; columns (3)–(4): Tobit. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Thus far we have established the extent to which beliefs differed between Southerners and Northerners. Yet, it is not clear whether differences also exist in the accuracy of their beliefs. [Uslaner \(2002\)](#) finds that optimism is a characteristic trait of high-trusting people in the sample he analyses. [Yamagishi \(2007\)](#) discusses the conjecture that trusting people may be no more than gullible people, inclined to erroneously put faith in others. We now aim to analyze the extent to which Southerners are indeed more pessimistic and less accurate in their beliefs than others.

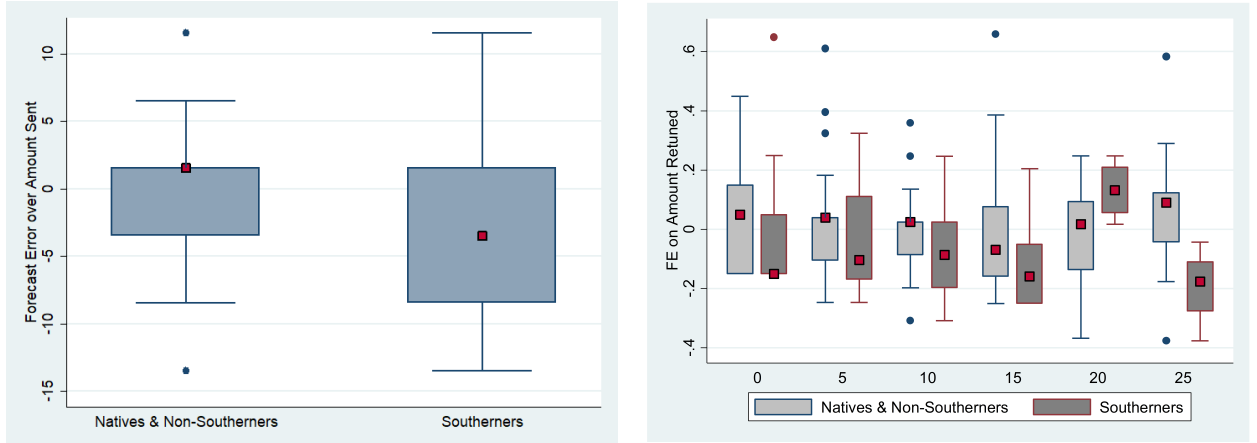
We construct a set of measures of forecast errors (FE) given by the difference between a subject’s expectations over the counterpart’s action and the average behavior actually observed in the experiment for the corresponding action. That is, we define FE for an agent i with respect to action k as $FE_i^k = E_i(x_k) - \bar{x}_k$, where $E_i(x_k)$ denotes subject i ’s expectation over a certain action k , $\bar{x}_k$ is the average value of action k observed in the experiment,¹² and FE is the forecast error. In the rest of the analysis, we define as “optimists” (“pessimists”)

¹²For subjects involved in the out-group treatment, we take a weighted average of actions by members and non-members. The weights reflect the actual relative number of association members over the total population in the province of Parma. According to ISTAT (2011), 11.21% of Parma residents are active voluntary members of some associations.

people having an $FE > 0$ ($FE < 0$).¹³ We also consider the absolute value of FE, which gives the magnitude of the error regardless of its sign.

The analysis reveals a significant bias towards pessimism for Southerners. The number of pessimists among Southern migrants is significantly higher than those observed in our sample of natives and non-Southern migrants (69% vs 47% using forecasts on amounts sent, 66% vs 46% using forecasts on amount returned; test of proportions, $p < 0.01$). Further, southern migrants make sizable errors in that they under-estimate others’ trust (4.20€ more pessimistic; Mann–Whitney test, $p < 0.001$; Figure 1a) and are generally more pessimistic than Natives and Non-Southerners together in estimating others’ trustworthiness (Figure 1b; Mann–Whitney test on average amounts returned, $p < 0.01$).

Result 2: Southerners hold significantly more pessimistic expectations, both for return rates and sending rates.



(a) Forecast errors over Amount sent by region of origin (b) Forecast errors over Return rate per each possible amount sent, by region of origin

Figure 1: Box plots for error forecasts over Amount sent (Panel a) and Return rate (Panel b) by region of origin. Squared points report average values. Category “Natives & Non-Southerners” pools together natives from Emilia Romagna and non-southern migrants.

3.2 The role of beliefs in explaining the level of trust and trustworthiness

Trust and Trustworthiness analysis. We now analyze the trust and trustworthiness levels. First, we notice that people from Italy’s South show lower levels of trust and trust-

¹³Our definition of “optimism” is based exclusively on the comparison between one’s own belief over others’ behavior and others’ actual behavior. We do not take into account alternative notions of “optimism” used in the literature such as over-confidence and expectation of favorable outcomes in random events (e.g. Deaux and Farris, 1977).

worthiness (Table 1). Trust among Southern migrants is significantly lower than that of Natives, broken down by groups. Migrants from Italy’s South stand out as being the group of people sending nothing with the highest frequency. Conversely, participants born in other areas are the group sending the largest amounts. As to the amount returned, Southerners return on average 30% less than both Natives and migrants from Italy’s North.

To analyze the determinants of the amount sent and of the return rate and to investigate the role of beliefs in explaining the behavior of people born in different parts of Italy, we perform an econometric analysis based on ordered logit and Random-effect Tobit models (eq. 5 in the SOM). On the other hand, the econometric analysis of the return rate is based on a Tobit model with random effects (eq. 6 in the SOM). The analysis on Senders’ actions (Table 3, model 1) shows the significantly lower levels among migrants from Italy’s South compared to Natives, even when accounting for an indicator variable for non-Southern migrants (*Non-Southern Migrants*). Both beliefs measures (*Amount sent exp* and *Return rate exp*) have a significant effect on Senders’ amount sent when considered on their own (Table 3, models 2–3). Hence, expecting the Receiver to return more leads to a higher amount sent. This supports the idea that the choice of how much to send was at least in part seen as a financial investment. The beliefs over what others would do in a similar situation also increase the amount sent, thus confirming the importance of social norms.¹⁴ Moreover, the introduction of beliefs into the regressions does change the significance of the variable *Southern Migrants*. As it can be seen in Table 3, *Southern Migrants* is significant when beliefs are not included in the regression. However, after controlling for beliefs, the difference between the amount sent by subjects born in the South of Italy and Natives disappears (see Table 3, columns 2–5). Lastly, similarly to Sapienza et al. (2013), we also introduce in the analysis a proxy for pro-sociality levels of individuals. This is taken by the decisions over how much to return to a Sender when individuals acted as Receivers. Cox (2004) shows that this variable is partly determined by altruism, partly by a desire to reciprocate Senders’ trust. It thus offers an estimate of important aspects of pro-social preferences. This variable averages the six return rates indicated by the subject when acting as Receivers (elicited through the strategy method; variable “Average return rate”). Such a variable is a strongly significant predictor of the amount sent ($p < 0.001$; Table 3, column 4). Similarly to when including belief variables, the introduction of *Average return rate* accounts for the effect of South, this latter becomes a non-significant predictor (Table 3, column 5).

In analyzing Receivers’ actions, we find a prominent role as well played by beliefs. People

¹⁴However, only the beliefs about amounts sent hold their significance in all model specifications. The variable *Return rate exp.* becomes not significant upon the inclusion of *Amount sent exp.*

from South of Italy send back significantly less than Natives, but this effect is less statistically significant once we control for the beliefs on the amount sent (*Amount Sent Exp*) and the beliefs on the amount returned (*Return rate exp*) (see Table 4).

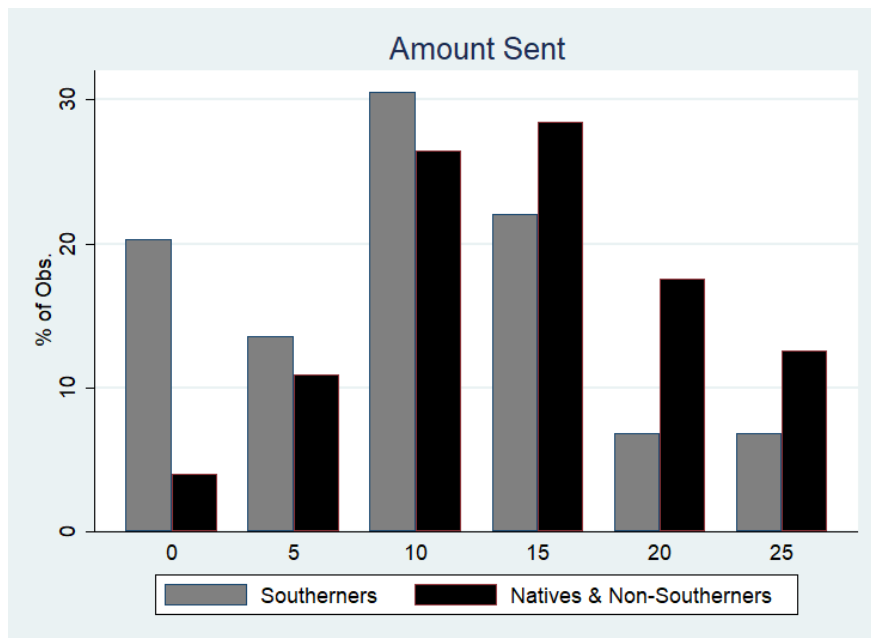


Figure 2: Histograms for Amount sent by region of origin. Southerners includes migrants from South of Italy. Natives & Non-Southerners includes both natives and migrants from regions other than South.

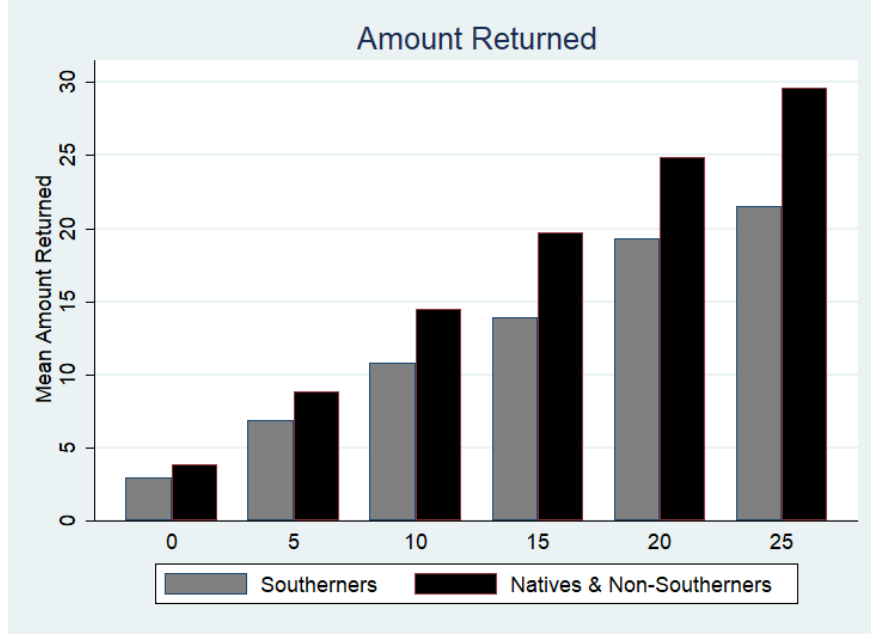


Figure 3: Average Return rates (amount returned) for each possible amount sent, broken down by region of origin. Southerners includes migrants from south of Italy. Natives & Non-Southerners includes both natives and migrants from regions other than South.

Mediation analyses. From our estimates, it emerges that the inclusion of at least one of the three variables accounting for beliefs and pro-sociality of individuals makes the effect of the indicator *Southern Migrants* no longer significant. It is reasonable to suspect that these variables mediate the effect of South. We run Sobel–Goodman mediation tests (Sobel, 1982) with bootstrapped standard errors¹⁵ to verify the extent to which these three measures (*Amount sent exp*, *Return rate exp*, *Average return rate*) are indeed mediators for *Southern Migrants* onto *Amount sent* (Figure 4) and *Return Rates* (Figure 5).

We initially consider the mediating effect of each variable separately, i.e., we test their mediation effect excluding the other two variables as covariates. In the case of trust onto others, the variable *Southern Migrants* has a significant indirect effect on *Amount Sent* for all three mediators (Figure 4, panels b–d; Aroian test; bootstrapped std. err. with 1000 repetitions), significant at $p < 0.001$ for *Amount sent exp* and at $p < 0.01$ for *Return rate exp* and *Average return rate*. The proportion of total effect that is mediated is larger for *Amount sent exp* (63%) than for *Return rate exp* (35%) and *Average return rate* (34%). If we include all three variables in the model as covariates, and we test for the mediating effect of each of them in turn, we find that only *Amount sent exp* has a significant mediating effect

¹⁵As remarked in Preacher and Hayes (2004), the bootstrapping method allows one to relax the normality assumption in the variables considered.

($\beta = -0.88$, $p = 0.036$; proportion of total effect mediated = 63%), while neither *Return rate exp* ($\beta = -0.15$; $p = 0.32$) nor *Average return rate* ($\beta = -0.28$; $p = 0.20$) have a significant indirect effect. We now analyze the role of risk aversion on senders' behavior. All regressions concerning the Senders' behavior include as a control variable a survey measure of willingness to take financial risks (variable "Risfin"). This has proved to be a reliable measure of risk aversion (Dohmen et al., 2011), and it has been demonstrated to correlate strongly with an experimental measure of risk aversion (Fehr et al., 2002; Falk et al., 2023). This variable positively affects the amount sent in the game (Table 3) meaning that the more the subjects are prone to take risks, the higher is the amount sent. However, it does not mediate for the effect of South, whose significance disappears only when belief measures are included in the regression (Figure 4).

We also perform the mediation analysis for the amounts returned, using the belief about the amount sent (*Amount Sent Exp.*) and the belief about return rates (*Return Rate Exp.*) as mediators (Figure 5). We explicitly do not explore the mediation effect of the actual *Amount Sent* because it has an impact on Return Rates by design, as participants play the role of sender first. The estimated mediation effect of the belief about the amount sent (*Amount Sent Exp.*) is higher (42%) than that of the belief about return rates (*Return Rate Exp.*) (30%; Figure 5, panels b–c). Similar results are obtained when both variables are included as covariates of the mediation analysis ($\beta = -0.02$; $p < 0.001$; proportion of mediated effect for *Amount Sent Exp.*: 34%; $\beta = -0.01$; $p < 0.01$; proportion of mediated effect of *Return rate Exp.*: 19%).

It seems clear that beliefs on both amounts sent and returned are key factor in explaining behavioral differences between Northerners and Southerners. We thus conclude:

Result 3: The lower trust and trustworthiness among Southerners in comparison to Northerners can be accounted for by their beliefs about others' behavior. The belief over the amount sent by Senders has the strongest mediating effect among the measures we used.

Table 3: Ordered logit regression on Amount Sent

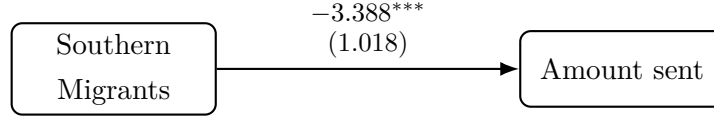
Dependent variable: Amount sent	(1)	(2)	(3)	(4)	(5)
Southern Migrants	-1.065*	-0.614	-0.762	-0.697	-0.369
	(0.425)	(0.442)	(0.428)	(0.459)	(0.465)
Non-Southern Migrants	-0.020	-0.088	0.002	0.014	-0.050
	(0.254)	(0.254)	(0.271)	(0.250)	(0.259)
Amount sent exp		0.235***			0.204***
		(0.025)			(0.027)
Return rate exp			3.663***		1.128
			(0.822)		(0.796)
Average Return rate				5.410***	3.298***
				(0.831)	(0.941)
Risfin	0.094	0.107*	0.078	0.130*	0.120*
	(0.053)	(0.049)	(0.051)	(0.055)	(0.051)
Demographics and additional controls	Yes	Yes	Yes	Yes	Yes
Observations	319	318	317	319	317

Standard errors in parentheses. 55 out of the 372 subjects were excluded because of missing values in the controls implemented. Results are unchanged also without controls (see SOM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

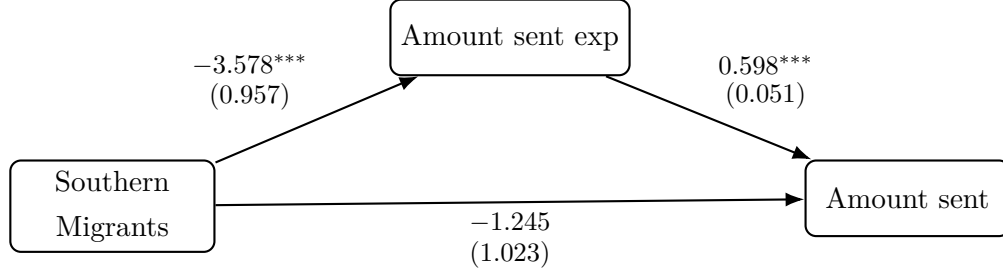
Table 4: Tobit regression on Return Rates

Dependent variable: Return Rate	(1)	(2)	(3)	(4)	(5)
Southern Migrants	−0.115*** (0.032)	−0.115*** (0.032)	−0.072* (0.031)	−0.084** (0.031)	−0.060 (0.030)
Non-Southern Migrants	−0.018 (0.025)	−0.018 (0.025)	−0.026 (0.023)	−0.016 (0.024)	−0.022 (0.023)
Amount Sent		0.029*** (0.002)			0.029*** (0.002)
Amount Sent Squared		−0.001*** (0.000)			−0.001*** (0.000)
Amount Sent Exp			0.011*** (0.002)		0.009*** (0.002)
Return rate exp				0.304*** (0.056)	0.199*** (0.058)
Constant	0.0883 (0.58)	−0.123 (−0.80)	0.00525 (0.04)	0.0392 (0.27)	−0.227 (−1.59)
Demographic and additional controls	Yes	Yes	Yes	Yes	Yes
Observations	1920	1920	1914	1908	1908

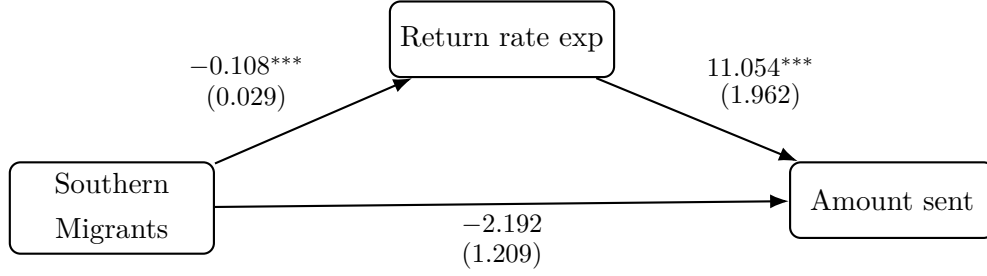
Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



(a) Total effect (no mediator)

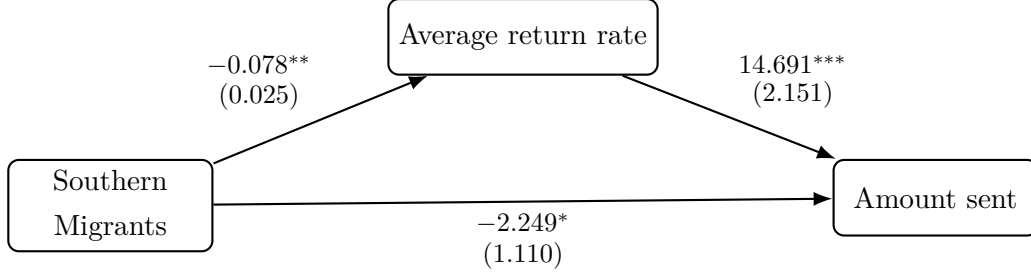


(b) Mediator: Amount sent exp. Direct effect $c' = -1.245$ (1.023, n.s.); indirect (mediated) effect $a \times b = -2.139^{***}$ (0.591); total $c = -3.384^{***}$ (1.019); % mediated = 63%.

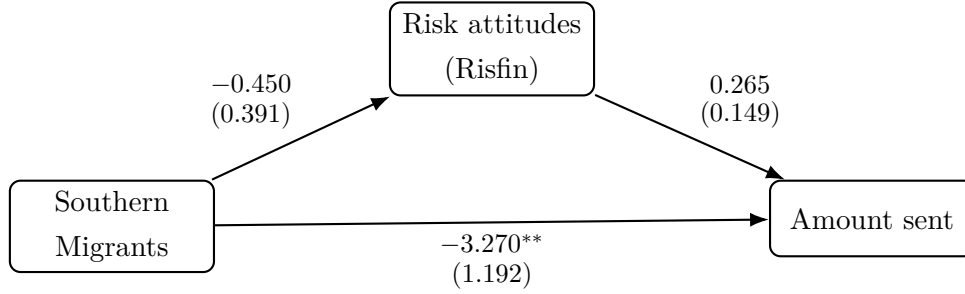


(c) Mediator: Return rate exp. Direct effect $c' = -2.192$ (1.209, n.s.); indirect (mediated) effect $a \times b = -1.191^{**}$ (0.421); total $c = -3.383^{***}$ (1.021); % mediated = 35%.

Figure 4: Mediating effect of Amount Sent Exp., Return Rate Exp., Avg. Amount Returned, and Risk attitudes on the relationship between Southern Migrants and Amount Sent (continued on next page). Path a : $X \rightarrow M$; path b : $M \rightarrow Y$; path c' : direct effect (c : total effect). Indirect and direct effects use bootstrapped std. err. (1000 reps); a , b , and total c use analytic std. err. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

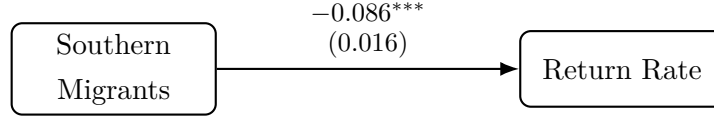


(d) Mediator: Average return rate. Direct effect $c' = -2.249^*$ (1.110); indirect (mediated) effect $a \times b = -1.140^{**}$ (0.365); total $c = -3.389^{***}$ (1.009); % mediated = 34%.

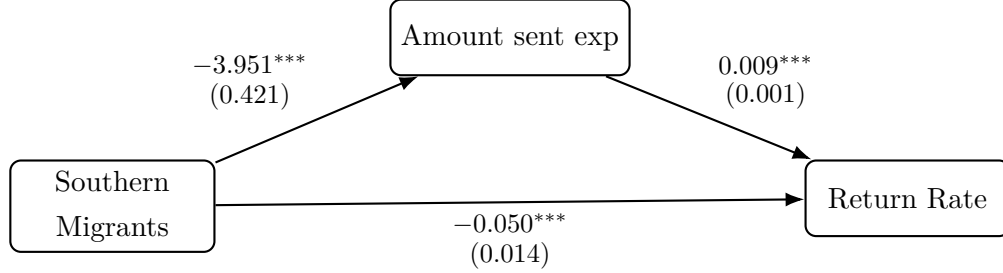


(e) Mediator: Risk attitudes (Risfin) — no mediation. Direct effect $c' = -3.270^{**}$ (1.192); indirect (mediated) effect $a \times b = -0.119$ (0.142, n.s.); total $c = -3.389^{***}$ (1.010); % mediated = 3.5%.

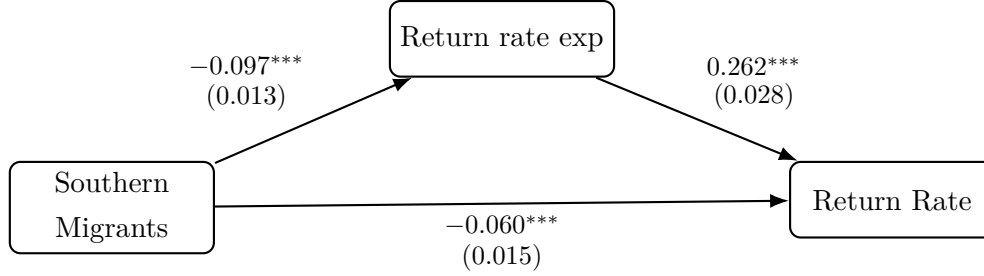
Figure 4: (continued) Mediating effect of Amount Sent Exp., Return Rate Exp., Avg. Amount Returned, and Risk attitudes on the relationship between Southern Migrants and Amount Sent. Path a : $X \rightarrow M$; path b : $M \rightarrow Y$; path c' : direct effect (c : total effect). Indirect and direct effects use bootstrapped std. err. (1000 reps); a , b , and total c use analytic std. err. $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$.



(a) Total effect (no mediator)



(b) Mediator: Amount sent exp. Direct effect $c' = -0.050^{***}$ (0.014); indirect (mediated) effect $a \times b = -0.036^{***}$ (0.005); total $c = -0.086^{***}$ (0.016); % mediated = 42%.



(c) Mediator: Return rate exp. Direct effect $c' = -0.060^{***}$ (0.015); indirect (mediated) effect $a \times b = -0.025^{***}$ (0.005); total $c = -0.086^{***}$ (0.016); % mediated = 30%.

Figure 5: Mediating effect of Amount Sent Exp. and Return Rate Exp. on the relationship between Southern Migrants and Return Rate. Path a : $X \rightarrow M$; path b : $M \rightarrow Y$; path c' : direct effect (c : total effect). Indirect and direct effects use bootstrapped std. err. (1000 reps); a , b , and total c use analytic std. err. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

3.3 Convergence over time

We now aim at studying whether Southern migrants' behavior and beliefs converge over time towards that of non-southern migrants. To do so, we rely on a subset of our sample of migrants for which the data of the time spent in the province of Parma is available.¹⁶ The main characteristics of the subsample are reported in the SOM (Table S8). Overall, 89 subjects are included in such subsample, 41 of which from the south of Italy, while the remaining 48 being from both Italy's center and other northern cities. Subjects in this sub-sample do not differ in the amounts sent and returned on average in the Trust Game

¹⁶We re-contacted migrants upon the experiment to administer an additional questionnaire. Not all the participants of the experiment were reached, thus the data is available for 70% of our sample of migrants.

compared to those subjects who are not in the sub-sample (Mann–Whitney on: amount sent, $p > 0.10$; belief on amount sent and amount returned $p > 0.10$).¹⁷ Additionally, subjects in the sub-sample do not differ in their main demographics with respect to those missing.¹⁸ To analyze the behavior of senders and receivers, we complement the models used in the previous sections with an index variable accounting for subjects’ number of years spent in the province of Parma relative to their age (*Years in Parma*).¹⁹ The subsample includes subjects who just moved in the province (*Years in Parma* around 0) as well as subjects who moved in early stages of their lives (*Years in Parma* around 1). A non-parametric test reports no significant difference between Southerners and Non-Southerners in the time spent in the province (Mann–Whitney test, $p = 0.79$).

Table 5 shows the impact of the time of residence in the region of Parma on both amounts sent and returned. Table 5, columns 1–2 report the estimates of an ordered logit model on amounts sent in the TG (for the full model specifications, see Table S9 in the SOM). We indicate with *Years in Parma* $\times$ *Southern Migrants* the interaction between *Years in Parma* and *Southern Migrants*. While levels of trust and trustworthiness are significantly lower among southern migrants relative to non-southern migrants (Table 5, models 1 and 3), regression results show no evidence of the convergence of Southern migrants’ behavior. We find no significant effect of neither *Years in Parma* nor *Years in Parma* $\times$ *Southern Migrants* on amounts sent and returned (models 2 and 4).²⁰

We now pass on the convergence analysis of belief on trust and trustworthiness. Even if beliefs about trust differ across Southern and non-Southern migrants, model estimates do not highlight any convergence path among Southerners (Table 6, models 1–2). However, we do find evidence suggesting a convergence in beliefs about the amounts returned (Table 6, models 3–4). While non-Southerners migrants hold more optimistic beliefs relative to Southerners, we find an heterogenous effect of *Years in Parma* across groups (model 4).

¹⁷When looking at amounts returned individually, they differ in some of the amount returned (i.e., amounts returned given an amount sent of 15€ and 20€; $p = 0.03$ and $p = 0.002$, respectively).

¹⁸We tested whether immigrants included in the subsample display significant differences in the main variables used in our analyses relative to those subjects whose time of residence is not available in our dataset. We find no differences in subjects’ age and education, number of unemployed subjects, their attitude towards risk, health satisfaction, or frequency of religious service attendance (p -values from t -tests report evidence of no significant statistical difference at any conventional level; results available upon request).

¹⁹More formally, this is given by the ratio between the number of years spent in the province and the age of the subject. In this way we focus only on the amount of lifetime that an individual has spent in the province of Parma and compare subjects of different ages.

²⁰Furthermore, results from a linear combination test on the coefficients associated with *Years in Parma* from our models do not provide evidence of a significant effect of the time spent in the region at any conventional levels (results from the combination test *Years in Parma* + *Years in Parma* $\times$ *Southern Migrants* = 0: $b = 2.605$, $p = 0.217$, for amount sent; $b = 0.171$, $p = 0.261$, for amount returned).

Table 5: Convergence analysis of amounts sent and returned

	Amount sent		Return Rate	
	(1)	(2)	(3)	(4)
Southern Migrants	-1.038**	-2.248*	-0.099***	-0.162
	(0.350)	(1.129)	(0.028)	(0.086)
Years in Parma		-1.105		-0.070
		(1.229)		(0.100)
Years in Parma $\times$ Southern Migrants		3.711		0.241
		(2.257)		(0.175)
Constant			0.074	-0.267
			(0.117)	(0.249)
Demographic and additional controls	Yes	Yes	Yes	Yes
Observations	334	93	2010	564
<i>Two-sided linear combination test. Null: Yrs. in Parma (for Southerners) = 0</i>				
Coefficient		2.605		0.171
<i>p</i> -value		0.217		0.261

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6: Convergence analysis of beliefs on amounts sent and returned

	(1)	(2)	(3)	(4)
	Amount sent exp	Amount sent exp	Return rate exp	Return rate exp
Southern Migrants	-1.091***	-2.132*	-0.122***	-0.246**
	(0.272)	(0.945)	(0.032)	(0.092)
Years in Parma		-0.734		-0.263**
		(1.133)		(0.084)
Years in Parma $\times$ Southern Migrants		2.183		0.361*
		(2.038)		(0.175)
Constant			0.111	-0.520
			(0.120)	(0.268)
Demographic and additional controls	Yes	Yes	Yes	Yes
Observations	333	92	332	92
<i>Two-sided linear combination test. Null: Yrs. in Parma (for Southerners) = 0</i>				
Coefficient		1.449		0.098
<i>p</i> -value		0.472		0.555

Standard errors in parentheses. The bottom rows report the test on the linear combination $Yrs. in Parma + Yrs. in Parma \times Southern Migrants$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4 Discussion and conclusions

Our paper analyzes the effect of beliefs, some preferences measures connected with altruism and risk aversion on the behavior of people from different Italian regions in a TG experiment. The population of Southern Italy has been widely studied in the recent literature on the determinants and effect of trusting behavior. However, we carry out the first analysis combining an experiment involving subjects, from different Italian regions, of different age and education levels with the elicitation of beliefs.

[Bigoni et al. \(2016\)](#) conducted Trust Game experiments between people born and resident in different Northern and Southern Italian locations. They find that trust is between 18% and 42% higher in Italy’s Northern locations than in Southern locations. In our experiment, trust and trustworthiness by Southerners are about 29% and 26% lower, respectively, than levels of native subjects. Southerners’ tendency to trust – and repay trust – less than Northerners makes their payoffs significantly lower. We also find evidence that emigrants from Italy’s South preserve levels of low trust and trustworthiness irrespectively to the length of time they have been relocated. This evidence is supported by Southerners’ reluctance to adapt their beliefs over time.

The result we observe in our experiment captures a behavioral difference between Italy’s Southerners and Northerners. In a closely related paper, [Bigoni et al. \(2019\)](#) find that Southerners are generally expected to be less cooperative by both southern and northern populations. We show that such behavioral differences between Northerners and Southerners are rooted into divergent beliefs over others’ behavior. Southerners have significantly more pessimistic expectations over others’ pro-social attitudes. Beliefs account for observed differences for both trust and trustworthiness. This result is robust to several controls. In particular, the analysis reveals a crucial role for the belief over the amount sent by Senders in mediating the differences between Southerners’ and Northerners’ decisions in the game.

While our results shed light on the persistent difference between natives and migrants living in the same environment, we can only speculate on the importance of cultural factors in explaining such gap. [Acemoglu and Robinson \(2010\)](#) claim that institutions have an over-arching predominance over cultural differences as explanatory factors of economic performance. Others have claimed that the Italian North–South divide is an instance of a “poverty trap” due to economic structural conditions ([Beraldo, 2010](#); [Daniele and Malanima, 2013](#)). Some have even claimed that such differences can be accounted for by genetic factors, which translate into IQ differences ([Lynn, 2010](#)), though recent evidence showing that economic conditions strongly affect cognitive abilities ([Mani et al., 2013](#)) should make

one rather cautious on this inference.

Our conclusion that behavioral differences between Northerners and Southerners are essentially caused by the pessimistic beliefs of the latter is relevant for economic policy. On the basis of this finding, policies aiming at increasing generalized trust should target a change of beliefs rather than a change in preferences. In the light of recent experimental results indicating that beliefs are in general more malleable to change than preferences (Naef and Schunk, 2011; Thöni et al., 2012; Volk et al., 2012), one may think that the chances of success in overcoming Southern Italy’s low trust are higher in as much as they are caused by beliefs rather than preferences.

Nevertheless, we do not know how long it would take to reshape people’s beliefs. Subjects in our sample have lived in Northern Italy for at least one year before taking part in the experiment, and many of them have lived in Parma for some decades. From our convergence analysis, we can infer that beliefs have a high degree of persistence, consistently to previous empirical work (Tabellini, 2010).

Given the importance of trust and co-operation for the economic and institutional performance, our results on people from the South of Italy, a wide and populated area characterized by low growth rates, deserve particular attention, and pose interesting questions for further research. Even knowing that beliefs over others’ behavior have a crucial role in explaining the low propensity to trust of people from Southern Italy, the question is still open as to what is the most effective way to alter such beliefs in order to increase trust and trustworthiness.

References

- Daron Acemoglu and James A. Robinson. Why is Africa poor? *Economic History of Developing Regions*, 25(1):21–50, 2010.
- Alberto Alesina and Nicola Fuchs-Schündeln. Good-bye Lenin (or not?): The effect of communism on people’s preferences. *American Economic Review*, 97(4):1507–1528, 2007.
- Yann Algan and Pierre Cahuc. Trust, growth, and well-being: New evidence and policy implications. In Philippe Aghion and Steven N. Durlauf, editors, *Handbook of Economic Growth*, volume 2, chapter 2, pages 49–120. Elsevier, 2014.
- Giulia Andrighetto, Andrea Guido, Elena Polizzi, and Aron Szekely. What motivates giving in dictator games? estimating the role of social norms and preferences. Working paper, July 2, 2026, 2026.

- Kenneth J. Arrow. *The Limits of Organization*. Norton, New York, 1974.
- Delia Baldassarri. Market integration accounts for local variation in generalized altruism in a nationwide lost-letter experiment. *Proceedings of the National Academy of Sciences*, 117(6):2858–2863, 2020.
- Edward C. Banfield. *The Moral Basis of a Backward Society*. Free Press, Glencoe, IL, 1958.
- Sergio Beraldo. Do differences in IQ predict Italian north–south differences in income? a methodological critique to Lynn. *Intelligence*, 38(5):456–461, 2010.
- Joyce Berg, John Dickhaut, and Kevin McCabe. Trust, reciprocity and social history. *Games and Economic Behavior*, 10:122–142, 1995.
- Cristina Bicchieri. *The Grammar of Society: The Nature and Dynamics of Social Norms*. Cambridge University Press, Cambridge, 2005.
- Maria Bigoni, Stefania Bortolotti, Marco Casari, Diego Gambetta, and Francesca Pancotto. Amoral familism, social capital, or trust? the behavioural foundations of the Italian north–south divide. *Economic Journal*, 126(594):1318–1341, 2016.
- Maria Bigoni, Stefania Bortolotti, Marco Casari, and Diego Gambetta. At the root of the north–south cooperation gap in Italy: Preferences or beliefs? *Economic Journal*, 129(619):1139–1152, 2019.
- Iris Bohnet and Yael Baytelman. Institutions and trust: Implications for preferences, beliefs and behaviour. *Rationality and Society*, 19(1):99–135, 2007.
- Lisa Cameron, Nisvan Erkal, Lata Gangadharan, and Marina Zhang. Cultural integration: Experimental evidence of convergence in immigrants’ preferences. *Journal of Economic Behavior and Organization*, 111:38–58, 2015.
- James C. Cox. How to identify trust and reciprocity. *Games and Economic Behavior*, 46:260–281, 2004.
- James C. Cox and Wafa Hakim Orman. Trust and trustworthiness of immigrants and native-born Americans. *Journal of Behavioral and Experimental Economics*, 57:1–8, 2015.
- Vittorio Daniele and Paolo Malanima. Falling disparities and persisting dualism: Regional development and industrialisation in Italy, 1891–2001. *Investigaciones de Historia Económica — Economic History Research*, 10(3):165–176, 2013.
- Giacomo Degli Antoni and Gianluca Grimalda. Groups and trust: Experimental evidence on

- the Olson and Putnam hypotheses. *Journal of Behavioral and Experimental Economics*, 61:38–54, 2016.
- Giacomo Degli Antoni and Gianluca Grimalda. Is social capital bridging or bonding? evidence from a field experiment with association members. *Theory and Decision*, 97(1):167–190, 2024.
- Thomas Dohmen, Armin Falk, David Huffman, Uwe Sunde, Jürgen Schupp, and Gert G. Wagner. Individual risk attitudes: Measurement, determinants and behavioural consequences. *Journal of the European Economic Association*, 9(3):522–550, 2011.
- Armin Falk, Anke Becker, Thomas Dohmen, David Huffman, and Uwe Sunde. The preference survey module: A validated instrument for measuring risk, time, and social preferences. *Management Science*, 69(4):1935–1950, 2023.
- Ernst Fehr, Urs Fischbacher, Bernhard von Rosenblatt, Jürgen Schupp, and Gert G. Wagner. A nation-wide laboratory: Examining trust and trustworthiness by integrating behavioral experiments into representative surveys. *Schmollers Jahrbuch (Journal of Contextual Economics)*, (4):519–542, 2002.
- Urs Fischbacher, Simon Gächter, and Ernst Fehr. Are people conditionally cooperative? evidence from a public goods experiment. *Economics Letters*, 71(3):397–404, 2001.
- Pablo Guillén and Daniel Ji. Trust, discrimination and acculturation: Experimental evidence on Asian international and Australian domestic university students. *Journal of Socio-Economics*, 40(5):594–608, 2011.
- Luigi Guiso, Paola Sapienza, and Luigi Zingales. The role of social capital in financial development. *American Economic Review*, 94(3):526–556, 2004.
- Luigi Guiso, Paola Sapienza, and Luigi Zingales. Social capital as good culture. *Journal of the European Economic Association*, 6(2–3):295–320, 2008.
- Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, Herbert Gintis, Richard McElreath, Michael Alvard, Abigail Barr, Jean Ensminger, Natalie Smith Henrich, Kim Hill, Francisco Gil-White, Michael Gurven, Frank W. Marlowe, John Q. Patton, and David Tracer. ‘economic man’ in cross-cultural perspective: Behavioral experiments in 15 small-scale societies. *Behavioral and Brain Sciences*, 28(6):795–815, 2005.
- Stephen Knack and Philip Keefer. Does social capital have an economic payoff? a cross-country investigation. *Quarterly Journal of Economics*, 112:1251–1287, 1997.

- Richard Lynn. In Italy, north–south differences in IQ predict differences in income, education, infant mortality, stature, and literacy. *Intelligence*, 38(1):93–100, 2010.
- Anandi Mani, Sendhil Mullainathan, Eldar Shafir, and Jiaying Zhao. Poverty impedes cognitive function. *Science*, 341(6149):976–980, 2013.
- Michael Naef and Daniel Schunk. Once bitten, twice shy: On the causal effect of prior experiences on trusting behaviour. 2011.
- Kristopher J. Preacher and Andrew F. Hayes. SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior Research Methods, Instruments, & Computers*, 36(4):717–731, 2004.
- Robert D. Putnam, Robert Leonardi, and Raffaella Y. Nanetti. *Making Democracy Work: Civic Traditions in Modern Italy*. Princeton University Press, Princeton, 1993.
- Fabio Sabatini. Social capital and the quality of economic development. *Kyklos*, 61(3):466–499, 2008.
- Paola Sapienza, Anna Toldra-Simats, and Luigi Zingales. Understanding trust. *Economic Journal*, 123:1313–1332, 2013.
- Michael E. Sobel. Asymptotic confidence intervals for indirect effects in structural equation models. *Sociological Methodology*, 13:290–312, 1982.
- Guido Tabellini. Culture and institutions: Economic development in the regions of Europe. *Journal of the European Economic Association*, 8(4):677–716, 2010.
- Christian Thöni, Jean-Robert Tyran, and Erik Wengström. Microfoundations of social capital. *Journal of Public Economics*, 96:635–643, 2012.
- Eric M. Uslaner. *The Moral Foundations of Trust*. Cambridge University Press, Cambridge, 2002.
- Stefan Volk, Christian Thöni, and Winfried Ruigrok. Temporal stability and psychological foundations of cooperation preferences. *Journal of Economic Behavior and Organization*, 81:664–676, 2012.
- Toshio Yamagishi. The social exchange heuristic: A psychological mechanism that makes a system of generalised exchange self-sustaining. In Mark Radford, Susumu Ohnuma, and Toshio Yamagishi, editors, *Cultural and Ecological Foundation of the Mind*, pages 11–37. Hokkaido University Press, Sapporo, 2007.